KRUSHANG PRAJAPATI

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FPGA / VLSI Design • Embedded Systems

PROFESSIONAL SUMMARY

Third-year B.Tech Electronics and Communication Engineering student (SPI 7.3/10) focused on FPGA and digital design: RTL in Verilog HDL, FSM-based design, testbench verification and implementation on Xilinx Vivado (Basys3, Artix-7). Won 1st place among 22 teams at FPGA Design Challenge 2.0 for a real-time image processing system, and built a cache controller, UART/webcam streaming pipeline, ESP32-CAM IoT systems and offline AI tools. Seeking Frontend VLSI and FPGA Applications Engineering internships.

EDUCATION

Sarvajanik College of Engineering & Technology, Surat — B.Tech Electronics and Communication Engineering *2024 – 2028*
Sarvajanik University | SPI: 7.3 / 10 (Sem 4)

BhagwanMahavir School, Surat — HSC (CBSE): 64% *2024*

B.S Patel Primary School, Bilimora — SSC (CBSE): 79% *2022*

PROJECTS

FPGA-Based Real-Time Image Processing System *Academic Project*
Verilog HDL · Xilinx Vivado · Basys3 (Artix-7) · FSM Design
github.com/Krushang004/FPGA-Image-Processing

- Designed and implemented a real-time image processing system on the Basys3 (Artix-7) FPGA in Verilog HDL, applying 8 image filters in hardware.
- Won 1st place among 22 teams at the FPGA Design Challenge 2.0, SCET, Surat.

UART Communication & Live Webcam Streaming (in progress) *Academic Project*
Verilog HDL · Python (OpenCV) · Icarus Verilog · Basys3 FPGA

- Built UART transmitter and receiver modules in Verilog with testbenches on the Basys3 (Artix-7).
- Building a live webcam-to-FPGA pipeline: custom Verilog frame-sync receiver FSM, dual-port BRAM frame buffer and a Python/OpenCV sender; Phase 1 verified with testbenches in Icarus Verilog.

Direct-Mapped Write-Back Cache Controller *Academic Project*
Verilog HDL · Xilinx Vivado · Basys3 FPGA · FSM Design
github.com/Krushang004/Cache-controller

- Designed a direct-mapped, write-back, write-allocate cache controller in Verilog HDL: 5-state FSM (IDLE, COMPARE, WRITE_BACK, MEM_FETCH, UPDATE), 256 lines, 4-word line size.
- Verified with a testbench; added a Basys3 top-level (switches/LEDs) to demonstrate hits, misses and write-backs.

FPGA Digital Design Implementations & XOR Encryption Engine *Academic Project*
Verilog HDL · Xilinx Vivado · Basys3 FPGA

- Implemented Sequence Detector (Moore & Mealy FSM), Sequence Generator, arithmetic units (Adder / Subtractor), Flip-Flops, MUX, and DSD modules in Verilog HDL — validated on FPGA.
- Built parameterised encryption / decryption hardware with configurable key width and key-locked decryption logic — validated on FPGA.

ESP32-CAM AI Motion Detection & Monitoring System *Academic Project*
ESP32-CAM · Firebase Realtime Database · AI Motion Detection

- Developed a smart surveillance system using ESP32-CAM with an AI-based motion detection model, sending alerts and data to a Firebase realtime dashboard for remote monitoring.

Smart Classroom IoT Dashboard *Personal Project*
Next.js · TypeScript · Firebase · ESP32-CAM
github.com/Krushang004/Smart-Classroom-IoT-Dashboard | Live demo

- Built a monitoring dashboard showing real-time temperature and occupancy data with device control.
- Integrated the ESP32-CAM live feed and AI motion-detection logs using Firebase Realtime Database.

ATmega32 AVR Assembly Programs *Academic Project*

AVR Assembly · ATmega32 · Proteus

- Wrote assembly programs for UART communication, 7-segment display control, delay loops and LED blinking; simulated in Proteus.

Offline AI Coding Assistant *Personal Project*

Python · Llama · Local LLM

github.com/Krushang004/Offline-AI-Assistant

- Built an offline, Llama-based assistant in Python that helps developers with code while keeping all data local and secure.

Python HTTP Honeypot *Personal Project*

Python · HTTP · Network Security

github.com/Krushang004/Honey-pot

- Built a lightweight honeypot that mimics a vulnerable web server and logs all incoming requests to detect unauthorized access, for security research.

Mindscape — Mental Wellness & Focus Tracker *Personal Project*

TypeScript · Android

github.com/Krushang004/Mindscape

- Built an app for daily mood logging, intention setting and reflection, to support mental wellness and focus.

TECHNICAL SKILLS

HDL & FPGA: Verilog HDL, Xilinx Vivado, Icarus Verilog, Basys3 (Artix-7), FSM Design, Testbench Verification, Block RAM, UART

Embedded & IoT: ESP32, ESP32-CAM, ATmega32 (AVR), Arduino, Sensor Integration, Relay Modules, IoT Dashboards

Languages: C, Python, AVR Assembly, MATLAB, TypeScript, JavaScript

Electronics & Simulation: Analog & Digital Electronics, Op-Amps, Circuit Design & Prototyping, LTspice, Proteus, MATLAB Simulink

AI / ML & Vision: Computer Vision, Motion Detection, ML Fundamentals, Offline LLM (Llama)

Web & Cloud: React, Next.js, Tailwind CSS, Node.js, Firebase (Realtime Database, Authentication)

Tools: Linux (Arch Linux & Ubuntu), Git, GitHub, Command Line Tools

LEADERSHIP & EXTRACURRICULAR

Google Student Ambassador *August 2025 – January 2026*

- Explored Gemini AI capabilities and conducted sessions training college students to use AI tools for academic and professional workflows.

IEEE Student Member | T&P Cell Volunteer *2024 – 2025*

- Volunteered with the Training & Placement cell and took part in IEEE student activities at SCET.

Event Volunteer

- Volunteer — Cyber Workshop, Sarvajanik University, Surat.
- Volunteer — SAW 3.0 Pre-Symposium 2025, SCET.
- Participant — International Press, SUMUN 2025, Sarvajanik University.

ACHIEVEMENTS

- 1st Prize — FPGA Design Challenge 2.0 (22 teams), SCET, Surat
- 3rd Prize — Networking using Cisco Packet Tracer, SCET, Surat
- Winner — Chess Tournament, SCET, Surat